

Fares Mohamed

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Robotics and Embedded AI Engineer | Control Systems, Intelligent Interfaces, and Applied Automation

Summary

Senior AI student at Alexandria University building robotics-aware software, automation pipelines, and intelligent control interfaces. Experienced in Arduino and ESP32 systems, Python-based tooling, FK/IK-assisted motion workflows, and AI-driven educational and robotics applications.

Experience

Armstrong

AI and Automation Engineer

Jul 2024 – Present

Alexandria, Egypt

- Built AI-assisted automation pipelines for the Digital Egypt Cubs Initiative (DECI), supporting scalable technical content generation for education programs.
- Integrated scripts, visuals, and structured outputs into repeatable workflows for faster delivery and consistent quality.

Engineering for Kids

Technical Instructor

Jun 2023 – Present

Alexandria, Egypt

- Taught robotics, Arduino, circuits, AI, Python, and C++ through hands-on STEM projects for students aged 11–16.

O-Academy

Coding Instructor

Feb 2025 – Present

Remote

- Conduct live coding sessions across Scratch, Roblox Studio, Python, and MIT App Inventor for U.S.-based learners.

Projects

DUM-E: ESP32 Robotic Arm Platform

Python, ESP32, Streamlit, Robotics

- Built a robotic arm platform combining ESP32 firmware, a Streamlit control dashboard, and ROS2-style interfaces for calibration, control, and execution.
- Implemented controller mapping, sequence recording, FK/IK-assisted path planning, and guarded pick-and-place workflows.

VisionDeck CV Suite

Python, OpenCV, MediaPipe, YOLO, Streamlit

- Developed a computer vision dashboard for face detection, hand gesture recognition, and object detection with webcam and Android IP camera support.

Robotics and Embedded Systems Builds

Arduino, C++, Python

- Designed and programmed Arduino-based robots and embedded builds for obstacle avoidance, sensor integration, and automation workflows.

Education

Alexandria University, Faculty of Computer and Data Science

B.Sc. in Computer Science (Intelligent Systems)

2022 – 2026

Alexandria, Egypt

Leadership

ElCoder

Head of AI Committee

Alexandria, Egypt

- Led student workshops and technical sessions on AI and ML fundamentals.

Skills

Languages and Tools: Python, C++, Arduino, ESP32, Streamlit, Docker, n8n

Robotics and Embedded: FK/IK workflows, motion planning, controller mapping, robotic interfaces, embedded prototyping

AI and Systems: Computer vision, LLM systems, AI pipelines, automation workflows, multimodal systems

Certifications

Machine Learning Specialization | Deep Learning Specialization | Google Cloud Big Data and Machine Learning Fundamentals